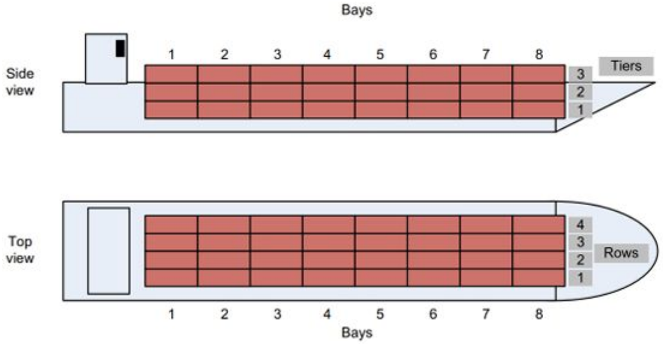
CARGO SHIP: PROBLEM DEFINITION AND DATA



GOALS:

- ❖ Provide an even and well-balanced solution.
- ❖ Ensure easy unloading solution.
- ❖ Maximize number of containers.

CONSTRAINTS:

- ❖ One container per cell (no overlapping).
- ❖ No floating containers (structure integrity).
- ❖ Acceptable weight difference when stacking (if $w_i > w_j$, then $w_i - w_j < \delta w$).

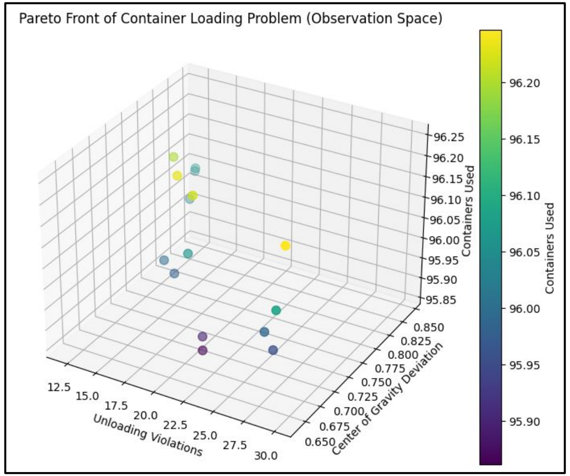
DATASET: simulates realistic loading scenarios with varied weights and unloading priorities:

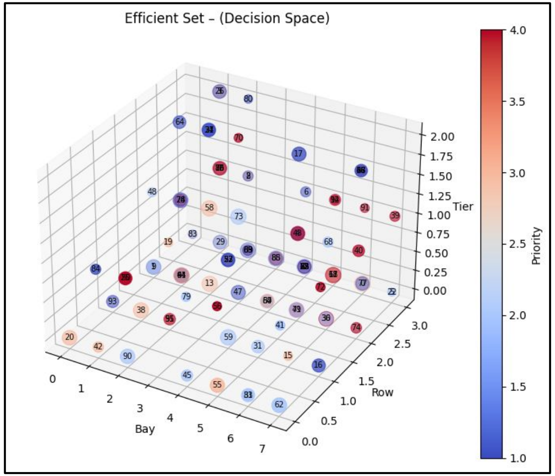
- ❖ 96 containers are generated to simulate a full-capacity scenario.
- ❖ Each container is assigned a random weight between 5 and 20 kg.
- ❖ Priorities (1 to 4) are randomly assigned to represent unloading urgency.

CARGO SHIP: ALGORITHMIC APPROACH: NSGA-II

- **NSGA-II** (from `pymoo`) solves a multi-objective container placement problem.
- Maintains a population, evolving solutions via **crossover** and **mutation**.
- Uses **Pareto dominance** and **crowding distance** for selection, promoting both quality and diversity.
- Optimizes:
  - **Unloading order** violations (minimize)
  - **Center-of-gravity** deviation (minimize)
  - **Space utilization** (maximize)
- Enforces constraints on **unique placement**, **stacking structure** and **weight limits**.

CARGO SHIP: RESULTS





- ❖ Filtered subset of Pareto solutions that balance objectives well.
- ❖ Identified 15 Pareto-optimal solutions.
- ❖ Focuses on minimizing CoG deviation and violations simultaneously.
- ❖ All solutions use 96 containers, but differ in unloading violations and center of gravity (CoG) deviation.

- ❖ Efficient set plot should show container placement across bay, row and tier.
- ❖ Each point in the efficient set plot represents a container.
- ❖ The size of a point reflects the container's weight and its color indicates the container's priority level.

SUMMARY

KEY INSIGHTS AND IMPLICATIONS:

- ❖ **Strict unloading order**: Successfully placed all 96 containers while minimizing unloading order violations related to destination port priorities — only a small number of violations (29 container pairs) remained due to trade-offs with space utilization and structural feasibility.

- ❖ **Perfect weight balance**: Achieved a highly balanced container layout with a minimal center-of-gravity deviation of 0.6467, placing the solution among the top ~10–15% in terms of weight distribution.

- ❖ **Algorithm flexibility**: The **NSGA-II** approach effectively explores **Pareto-optimal solutions**, allowing decision-makers to choose based on priorities (i.e. **stability** vs **speed of unloading**)

Shipping companies can use this model to **reduce costs** and **speed up** port operations.